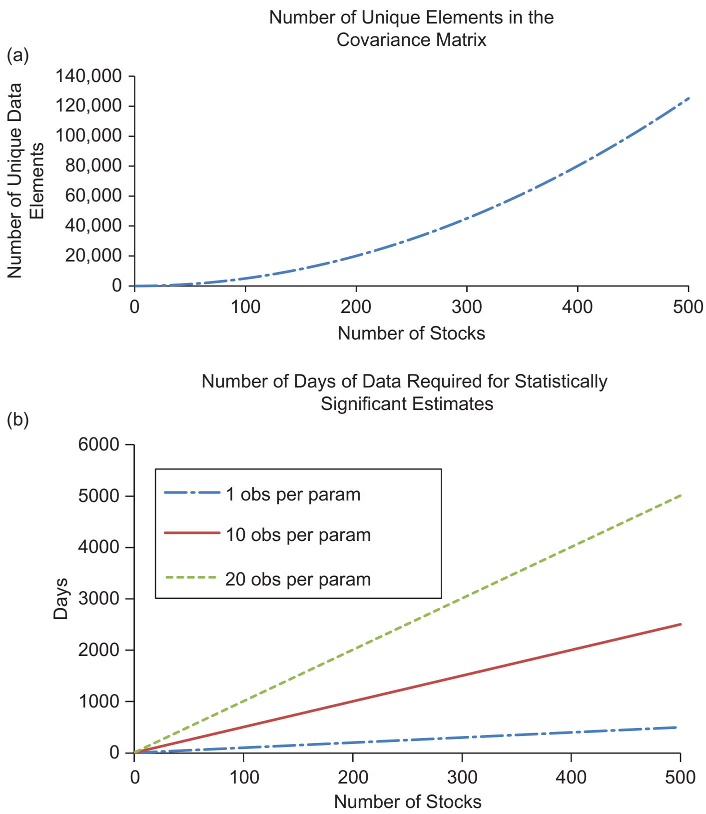

■ **Figure 6.8** Errors Associated with Estimating Stock Price Movement Covariance

The last point to make is that for a global covariance matrix with a global universe of over 50,000 companies (at least 50,000!) there would be over 1.25 billion unique parameters and we would need a historical prices series of over 100 years! Think about how much has changed in just the last 10 years, let alone 100 years.

FACTOR MODELS

Factor models address the two deficiencies we encountered when using historical market data to compute covariance and correlation. First, these models do not require the large quantity of historical observations that are needed for the sample covariance approach in order to provide accurate risk estimates. Second, factor models use a set of common explanatory factors across all stocks and comparisons are made to these factors